

evaluate(): average over N seeded replications

Scenario
(forces, terrain, objective)

seed

design params
(sweep knobs)

run_mission(scenario, seed, params) → metrics · pure function

build world (terrain rasters: speed, cover, conceal) + entities (SoA arrays)

**C2
tasking**

planning
(routes,
formation)

motion
(unicycle,
terrain)

sensing
(shared SA,
relay)

engagement
(aimed fire,
Pk)

for k in steps

read

fixed order, $dt = 1$ s

write

state: SoA entity arrays (x, y, heading, hp, seen, control_quality, ...) + terrain rasters

mission KPIs: success, blue/red losses, time-to-objective, detection coverage

polarisopt DOE / BO → SLURM (Hopper)